

Key differences between Ethereum, Hyperledger and R3 Corda

Features	Ethereum	Hyperledger	R3 Corda
Governance	Ethereum developers	Linux Foundation	R3 consortium
Smart contract programming	Solidity	Go, Java	Kotlin, Java
Consensus algorithm	Proof-of-work	Pluggable framework	Pluggable framework
Ledger type	Permissionless	Permissioned	Permissioned
Currency	Powered by Ether native currency	Does not have an inbuilt native cryptocurrency. However, this can be created using chain code	Does not have an inbuilt native cryptocurrency
Use cases	Popular with generalised applications and mostly used for business-to-consumer operations	Mostly used for business-to-business and enterprise applications	Preferred for financial services applications

be compatible with the Bitcoin network and thereby creates new cryptocurrency networks (e.g., Litecoin or Zcash).

- Facilitates transparency by requiring new entries to include a proof-of-work.
- It enhances transparency and offers new possibilities to be tapped.

Platforms for building blockchain technology

Some of the most popular blockchain frameworks include the following.

- **Ethereum:** This is an open source distributed cloud platform that enables developers to build and deploy decentralised applications. Ether is the built-in cryptocurrency of Ethereum that works on the public blockchain network. Ethereum is the first and the most popular smart contract platform that enables users to define smart contracts in the form of code.
- **Hyperledger:** This is an open source, modular architecture targeted at businesses that aim to streamline their processes leveraging blockchain technology. Ethereum is an open and distributed platform on which it is impossible to hide a transaction from anyone. Hyperledger, on the other hand, offers this flexibility.

- **R3 Corda:** This has a unique approach to data distribution, whereby the data is shared only between the authorised participants and not the entire network. Since it operates in a permission mode, it enhances privacy and offers fine-grained access control to digital records.

Industry use cases and trends of blockchain

- **Healthcare:** Different healthcare service providers use their own database systems (private), which contain crucial information about their patients. This is scattered and inaccessible (to other providers) and is very dangerous in an industry in which even a few extra seconds taken to obtain critical information could impact a patient's life.

Blockchain makes healthcare-related data easily accessible amongst all providers, which leads to better and faster treatment.

- **Insurance claims:** Blockchain technology can be applied to the insurance sector. Through a smart contract, customers' and insurers' claims can be managed in a transparent and secure manner.
- **Media:** Media companies have started to adopt blockchain technology to eliminate fraud, reduce costs and even protect intellectual property (IP) rights on content like original music.
- **Energy:** Another potential application of blockchain technology includes the execution of energy supply transactions to provide the basis for metering, billing and clearing processes. 

References

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- [2] <https://bitcoin.org/bitcoin.pdf>
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